

Omar Ahmed Fahmy

PHY & DSP Algorithm Engineer

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Professional Summary

Cross-disciplinary PHY & DSP Algorithm Engineer bridging system-level modeling and digital hardware implementation. Develop floating- and fixed-point models of low-PHY/DFE signal chains with realistic impairment modeling to guide algorithm and architecture choices. Assess DSP algorithms and micro-architectures against latency, throughput, and PPA constraints, then implement fixed-point designs in clean C for embedded software paths or synthesizable SystemVerilog with pipelined datapaths, verification, and FPGA prototyping on industry-standard toolchains.

Technical Skills

HDL, Programming, Scripting: MATLAB, Verilog / SystemVerilog, C, Python, TCL

DSP & PHY Processing: Floating- and Fixed-Point Modeling, Algorithm–Architecture Trade-off Analysis, Adaptive Filtering, Digital Pre-Distortion (DPD), FFT, CORDIC, QR Decomposition, Timing Synchronization, Curve Fitting

RTL Design & Verification: Microarchitecture Design, Pipelined Fixed-Point Datapaths, Clock Domain Crossing (CDC), Functional Verification, Low-Power Design, Code Coverage, Linting, Design For Test (DFT)

Tools: QuestaSim, SpyGlass, Design Compiler, Vivado

Professional Experience

PHY & DSP Algorithm Engineer – Analog Devices (Egypt) Sep 2023 – Present

- Built floating-/fixed-point PHY and DFE models incorporating analog/RF impairments and benchmarked performance (BER/BLER, EVM, ACLR) to drive algorithm and architecture decisions across teams
- Assessed DSP algorithm candidates and evaluated algorithm–architecture trade-offs to meet latency, throughput, and PPA constraints
- Implemented fixed-point DSP algorithms in clean C for firmware integration, ensuring bit-true alignment with system-level models
- Translated fixed-point DSP algorithms into synthesizable SystemVerilog with deeply pipelined datapaths and executed functional verification, linting, and code coverage
- Prototyped on AMD FPGAs using Vivado and on-chip debug (ILA/logic analyzer), validating functionality in the lab
- Led architecture analysis, technology porting, and SoC integration definition for a 12-GSPs 16-slice time-interleaved high-speed ADC digital subsystem — migrating from TSMC 16nm to GF 22FDX
- Performed focused link-level coexistence studies for 5G NR–Wi-Fi systems to quantify interference impacts on BER/BLER and throughput across deployment configurations

Quality Assurance Engineer – Siemens EDA (Egypt) Apr 2023 – Sep 2023

- Developed and maintained automated test scripts for Visualizer GUI features, contributing to ongoing regression coverage
- Investigated and resolved Jira-reported issues by reproducing tool behaviors, isolating root causes, and validating fixes with development engineers

Internships

- **Digital Design Intern – Si-Vision (2022):** UART-APB bridge design & verification; lint, CDC/RDC with Synopsys SpyGlass
- **Digital Design Intern – ONE Lab (2022):** accelerator modeling in C/C++ and Vivado HLS, RTL generation, FPGA validation
- **Digital Design Intern – PyramidTech LLC (2022):** FPGA project development using VHDL, simulation, timing analysis, and bring-up
- **Analog IC Design Intern – ITI / Dr. Hesham Omran (2021):** learned and applied advanced analog IC design topics including gm/ID design methodology
- **Embedded Software Intern – NilePreneurs (2020):** contributed to a microcontroller-based automotive cruise-control prototype

Key Engineering Projects

5G Low-PHY / DPD System-Level Modeling & Parameter Optimization

- Built a full floating-point MATLAB system model of the 5G NR transmit chain (NR generator, CFR, QEC, DPD, DAC/PA/ADC/VGA), incorporating realistic analog/RF impairments (I/Q imbalance, IMDs, delay, noise) to study end-to-end transmitter behavior
- Explored key architectural parameters — NR signal bandwidth, QAM order, subcarrier spacing, oversampling ratio, CFR PAPR target, DPD degree, DPD memory depth, DAC/ADC ENOB, VGA gain planning — through extensive parameter sweeps
- Ran comprehensive performance analyses (EVM/ACLR vs Pout, PAPR vs Pout, EVM vs DPD iterations), exposing system bottlenecks tied to quantization, CFR clipping, PA compression, and DPD update dynamics
- Delivered a validated system-level reference model used as the foundation for deeper DPD algorithmic design and hardware prototyping

Digital Pre-Distortion (DPD) Architecture, Algorithms & FPGA Prototyping

- Performed a rigorous comparison of DLA, ILA, and iDLA DPD architectures, evaluating performance, convergence speed, latency, and hardware implementation cost
- Designed the full DPD processing chain — normalizer, synchronizer, coefficient estimator, actuator—selecting algorithms through detailed latency/throughput/PPA trade-off analysis
- Evaluated multiple algorithmic options for each sub-block, including time vs frequency correlation for synchronization, FFT algorithm families, sinc vs Lagrange fractional-delay filters, and LU vs QR decomposition for coefficient estimation — ultimately selecting a hardware-efficient QR approach using Givens rotations implemented with CORDIC
- Performed fixed-point MATLAB analysis across all DPD sub-blocks optimizing word-lengths for efficient RTL implementation
- Implemented the whole DPD system in synthesizable SystemVerilog with pipelined datapaths, achieving bit-true alignment with MATLAB reference models
- Prototyped, validated, and debugged the entire DPD system on AMD UltraScale+ FPGA using on-chip ILA/logic analyzers for in-lab testing

32-Channel Beamformer Calibration (System Modeling, Algorithms & Embedded Firmware)

- Built a MATLAB end-to-end model of a 32-channel beamformer chip transmit/receive lineup to precisely select calibration-signal parameters and system operating conditions for reliable on-chip gain/phase and IMs measurements

- Designed two calibration algorithms for long-term PA aging compensation, including *Gain & Phase Calibration* via single-tone correlation and *IMs Estimation* via dual-tone excitation with four correlations to extract IMD-to-main-tone ratios
- Performed fixed-point MATLAB analysis to size all signals targeting ARM Cortex execution, then implemented both calibration algorithms as optimized C firmware, minimizing runtime and memory footprint and verifying correctness against MATLAB reference vectors

12-GSPs ADC Subsystem Integration & Cross-Node Technology Migration

- Led the end-to-end technology migration of a 12 GSPs 16-slice time-interleaved ADC digital subsystem — including its Cortex-M4 processor wrapper — from TSMC 16nm to GF 22FDX, replacing all standard cells, SRAMs, and IO primitives

Defined the integration architecture for deploying 8 ADC instances into a Cortex-M85 SoC, specifying the bus topology, arbitration protocol, and synchronized capture mechanism

Balanced Binary-Tree FFT (Algorithm & RTL Design)

- Studied FFT algorithm families (radix-k, radix-2^m, split-radix, mixed-radix) and selected the Balanced Binary-Tree Decomposition algorithm for area-efficient pipelined FFT processing
- Developed a parameterized MATLAB model capable of generating FFT architectures of arbitrary size with optimized multiplier count and minimized twiddle-factor memory
- Designed parameterized fixed-point RTL reflecting the balanced decomposition, enabling scalable FFT implementations with reduced hardware resources
- Verified functional correctness via simulation and bit-true comparison to reference model

Low-Power Multi-Clock Digital System (RTL to GDS)

- Applied a complete ASIC flow from RTL to GDS, including synthesis, timing closure, place-and-route, and physical sign-off
- Inserted DFT logic and performed functional, linting, and formal verification at RTL, post-synthesis, post-DFT, and post-PnR stages

UART-APB Bridge (RTL Design & Verification)

- Designed a protocol-accurate UART-APB bridge and validated correctness using self-checking testbenches and static verification (lint, CDC/RDC) using Synopsys SpyGlass

5G NR-Wi-Fi Coexistence Link Analysis

- Conducted focused link-level studies for coexisting radio systems, sweeping deployment parameters and quantifying BER/BLER/throughput impacts

Education

B.Sc. in Electronics & Electrical Communications Engineering
Cairo University (Egypt)

Sep 2018 – Jul 2023

- Graduated with **Excellent with Honors** (94%)
- Ranked **2nd** in class
- **Graduation Project:** 5G Low-PHY/DFE Modeling & FPGA Prototyping sponsored by Analog Devices

Certifications

- Functional Verification Diploma – Siemens EDA
- Advanced 5G Diploma – Siemens EDA

- 5G Development and Evolution Diploma – Siemens EDA
- Digital IC Design Diploma – Ali El Temsah

Awards & Honors

- **President's Recognition (2023):** honored by the President of Egypt as one of the nation's top-performing undergraduates
- **National Academic Excellence Award (2022):** honored by the President of Cairo University for ranking among the top students across all faculties
- **Top Thanaweya Amma Student Award (2018):** recognized by the Military Technical College for outstanding nationwide performance

Leadership & Activities

- **Graduation Projects Mentor – Analog Devices:** annually mentor graduation-project teams through weekly technical meetings covering PHY/DSP modeling, architecture trade-offs, and digital-hardware design
- **Internship Instructor – Analog Devices Digital Design Internship:** annually deliver technical learning sessions, define final projects, assess interns through individual presentations, and contribute to final certification decisions
- **Coursera Global Translator Community:** translated English courses into Arabic to expand access to online engineering education for Arabic-speaking learners
- **American Center in Cairo (ACC):** participated in weekly discussion sessions focused on communication topics
- **Al-Ahly SC Football Team – Team Captain:** led the team in 50+ matches